

Aim:**Part A — Practice Programs (Hour 1)**

Work these in your lab-NN/ folder before the project: do the first one or two in the lab; the last (For Advanced Learners) is for home.

1. Shape areas: an abstract Shape with abstract area(); subclasses Circle, Rectangle, Triangle. Put a mix in a Shape[] and loop once to print each area and the running total, tracking the largest — one loop handling every shape through polymorphism.

2. Payroll: an abstract Employee with abstract monthlySalary(); FullTime (fixed), PartTime (hours × rate) and Intern (stipend), each calling super for the shared name/id. Loop a mixed array to print each salary and the total; use instanceof to add a note only for interns.

3. (For Advanced Learners – home) Model a few media types that each compute a late fee differently and total the fees for a returned batch. Design the hierarchy and rules yourself.

Code Screenshot:

```
1  import java.util.Scanner;
2
3  abstract class Shape{
4      abstract double Area();
5  }
6
7  class Circle extends Shape{
8      private double radius;
9
10     public Circle(double radius){
11         this.radius = radius;
12     }
13
14     @Override
15     public double Area(){
16         return radius * radius * Math.PI;
17     }
18 }
19
20 class Rectangle extends Shape{
21     private double width, height;
22
23     public Rectangle(double width, double height){
24         this.width = width;
25         this.height = height;
26     }
27
28     @Override
29     public double Area(){
30         return width * height;
31     }
32 }
33
```

```
33
34 class Triangle extends Shape{
35     private double base, height;
36
37     public Triangle(double base, double height){
38         this.base = base;
39         this.height = height;
40     }
41
42     public double Area(){
43         return (base * height * 0.5);
44     }
45 }
46
47
48 public class ShapeArea{
    Run | Debug
49     public static void main(String[] args){
50         Scanner s = new Scanner(System.in);
51         double maxArea = 0;
52         System.out.print(s: "Enter number How many area you want to calculate: ");
53         int num = s.nextInt();
54         for(int i = 0; i < num; i++){
55             System.out.println();
56             System.out.println(x: "Circle --> 1");
57             System.out.println(x: "Rectangle --> 2");
58             System.out.println(x: "Triangle --> 3");
59             System.out.print(s: "Enter any number to calculate area: ");
60             int a = s.nextInt();
61
62             if (a == 1){
63                 System.out.print(s: "Enter Radius");
64                 double r = s.nextDouble();
65                 Shape c = new Circle(r);
66                 System.out.println("Area of circle is " + c.Area());
67                 if (maxArea < c.Area()){
```

```
68                 maxArea = c.Area();
69             }
70         }
71         else if(a == 2){
72             System.out.print(s: "Enter Width and Height respectively");
73             double w = s.nextDouble();
74             double h = s.nextDouble();
75             Shape rec = new Rectangle(w,h);
76             System.out.println("Area of Rectangle" + rec.Area());
77             if (maxArea < rec.Area()){
78                 maxArea = rec.Area();
79             }
80         }
```

```

81         else if(a == 3){
82             System.out.print(s: "Enter Base and Height respectively");
83             double b = s.nextDouble();
84             double h = s.nextDouble();
85             Shape tri = new Triangle(b,h);
86             System.out.println("Area of Triangle" + tri.Area());
87             if (maxArea < tri.Area()){
88                 maxArea = tri.Area();
89             }
90         }
91         else{
92             System.out.println(x: "Enter correct input");
93             i--;
94         }
95     }
96     System.out.println("Maximum Area is " + maxArea);
97
98     s.close();
99 }
100 }

```

Output Screenshot:

```

• jitpatel2010icloud.com@JitPatels-MacBook-Air Prac-5 % javac ShapeArea.java
• jitpatel2010icloud.com@JitPatels-MacBook-Air Prac-5 % java ShapeArea
Enter number How many area you want to calculate: 4

Circle --> 1
Rectangle --> 2
Triangle --> 3
Enter any number to calculate area: 1
Enter Radius24.6
Area of circle is 1901.1662102463995

Circle --> 1
Rectangle --> 2
Triangle --> 3
Enter any number to calculate area: 2
Enter Width and Height respectively45.3 40.6
Area of Rectangle1839.1799999999998

Circle --> 1
Rectangle --> 2
Triangle --> 3
Enter any number to calculate area: 4
Enter correct input

Circle --> 1
Rectangle --> 2
Triangle --> 3
Enter any number to calculate area: 3
Enter Base and Height respectively20 40.6
Area of Triangle406.0

Circle --> 1
Rectangle --> 2
Triangle --> 3
Enter any number to calculate area: 1
Enter Radius47
Area of circle is 6939.778171779853
Maximum Area is 6939.778171779853
○ jitpatel2010icloud.com@JitPatels-MacBook-Air Prac-5 %

```

Learning Outcome:

In this practical I understand the abstract class and override methods.